MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6528A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, medium Mu, series regulator

Outline --- See figure 1

Base --- B8-98 (glass-bonded mica)

Envelope --- T16

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ef V	Eb Vdc	Ec Vdc	Ib/a mAdc	Ehk v	Rk/k Ohms	Rg/g Ohms (Note 3)	Pp/a W	TE °C	tk ---	Alt ft
Maximum:	6.9	400	---	300	+300	---	50,000	30	250	---	Note 1
Minimum:	5.7	---	-300	---	-300	Note 2	500	---	---	Note 4	---
TEST CONDITIONS:	6.3	100	-4	---	---	0	500(min)	---	---	---	---

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification inspection							
(F) 1216	Base material insulating quality	Glass-bonded mica, zone 5 minimum	---	---	---	---	---	---
	Quality conformance inspection, part 1							
1231	Emission	Eb = Ec = 10 Vdc; Rk = 0 (see note 5)	0.65	II	Is	150	---	mAdc
1256	Electrode current (1) (anode)	See note 6	0.65	II	Ib	140	230	mAdc
1256	Electrode current (3) (anode)	Eb = 400 Vdc; Ec = -75 Vdc; Rk = 0 (see note 6)	0.65	II	Ib	---	3	mAdc
1266	Total grid current	Rg = 0.05 Meg (see note 6)	0.65	II	Ic	0	-2	µAdc
1336	Heater-cathode leakage	Ehk = +300 Vdc Ehk = -300 Vdc (see note 7)	0.65	II	Ihk	---	40	µAdc
1201	Short and discontinuity detection		0.65	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	4.75	5.25	A
1316	Amplification factor	See note 6	---	---	Mu	7	11	---
1211	Insulation of electrodes		4.0	S3	---	---	---	---
1261	Electrode voltage (anode)	Rk = 0; Ec = 0; Eb/Ib = 300 mAdc (see note 5)	---	---	Eb	---	125	Vdc
1256	Electrode current (2) (anode)	Eb = 60 Vdc; Ec = 0	---	---	Ib	135	---	mAdc
1306	Transconductance (1)	See note 6	---	---	Sm	29,000	45,000	µmhos
1306	Transconductance (2)	Ef = 5.7 V (see note 6)	---	---	ΔSm Ef	---	15	%
1201	Tap shorts	Ebb = 400 Vdc; Ec = -300 Vdc (see note 8)	---	---	---	---	---	---
1031	Variable-frequency vibration	Rp = 2,000 ohms; Ec/Ib = 75 mAdc; Ebb = 190 Vdc (see note 9)	---	---	Ep	---	300	mVac
1031	Vibration fatigue	50 to 500 to 50 Hz; 2.5 G; time of sweep 50 to 500 to 50 Hz = 3 minutes (min) to 15 minutes (max)	6.5	See note 10	---	---	---	---
---	Long duration shock	Shock = 11 ±2 ms; accel = 100 G peak (min); total impacts = 18, Ef = 6.3 Vac (see note 11)	15	S4	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
---	Post-shock and vibration-fatigue test end points	Electrode current (1) (anode)	---	---	Ib	130	---	mAdc
		Electrode current (grid)	---	---	Ic	---	-8	µAdc
		Transconductance (1)	---	---	Sm	23,000	---	µmhos
		Heater-cathode leakage Ehk = +300 Vdc Ehk = -300 Vdc	---	---	Ihk	---	100	µAdc
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1111	Base pin solder depth		---	---	---	---	---	---
1131	Metallic base sleeve quality		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; Rg = 0.5 Meg; TE = 250° ±25° C; Ehk = 300 Vdc (see note 12)	---	---	---	---	---	---
---	Life-test end points (1,000 hours)	Electrode current (1) (anode)	---	---	Ib	130	---	mAdc
		Electrode current(grid)	---	---	Ic	---	-8	µAdc
		Transconductance	---	---	Sm	23,000	---	µmhos
		Heater-cathode leakage Ehk = +300 Vdc Ehk = -300 Vdc	---	---	Ihk	---	100	µAdc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Minimum resistance per cathode leg shall be 10 ohms or that resistance necessary to provide 10 percent of the grid bias voltage, whichever is greater.
3. Maximum grid circuit resistance in regulator service or with fixed bias shall be 50,000 ohms. Maximum resistance for cathode bias operation shall be 500,000 ohms.
4. The cathode-conditioning time is 30 seconds minimum. This is especially necessary in regulator circuits where the anode of the amplifier tube ties back to the anode-side of the passing tube. With no clamping diode between grid and cathode in this type of circuit, the grid bias may swing highly positive causing the cathode to lose its ability to sustain current.
5. Each unit shall be read separately. Ground unit not under test.
6. Voltages shall be applied to both sections, but each section shall be read separately. Values specified herein are for individual sections, except for heater current. Readings shall be taken after tube has been preheated for at least 2 minutes under conditions: Ef = 6.3 V; Ebb = 190 V; Ec = 0; Rk = 200 ohms.
7. Connect 1k to 2k.
8. Test in circuit of figure 2. Manual tapping per method 1201 shall be employed. Any visible flashing of the neon lamps shall be considered an indication of a tap short. If a short indication is obtained, the tapping cycle shall be repeated two more times. If a short indication is again obtained, the tube shall be rejected.

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NOTES: -Continued

9. Tie 1k to 2k, 1g to 2g, and 1a to 2a.

10. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

11. The tubes shall be subjected to the specified acceleration in the X, Y, and Z axes, with six shocks in each axis. Tubes showing any permanent shorts, or more than one temporary short during the test, shall be rejected. Method of short testing is with thyratron controlled, short indicator as described in method 1041. The applied shock shall be an approximate half sine-wave motion with duration measured at the zero axis level. This is a destructive test.

12. The heater-to-cathode voltage shall be cycled 30 seconds positive and 30 seconds negative.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Navy -
Air Force - 99
DSA - ES

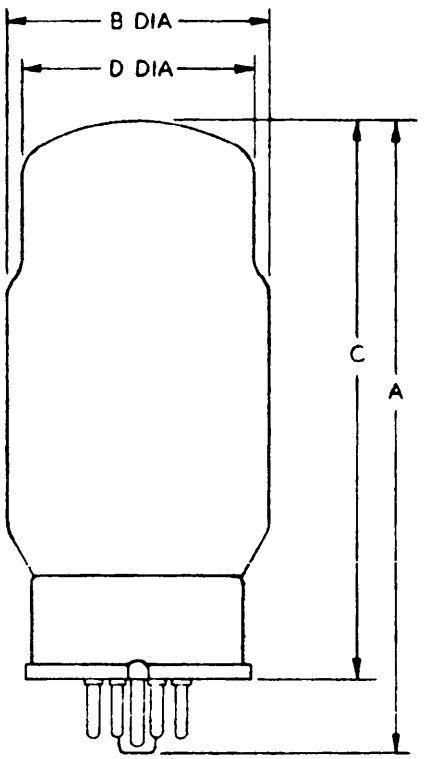
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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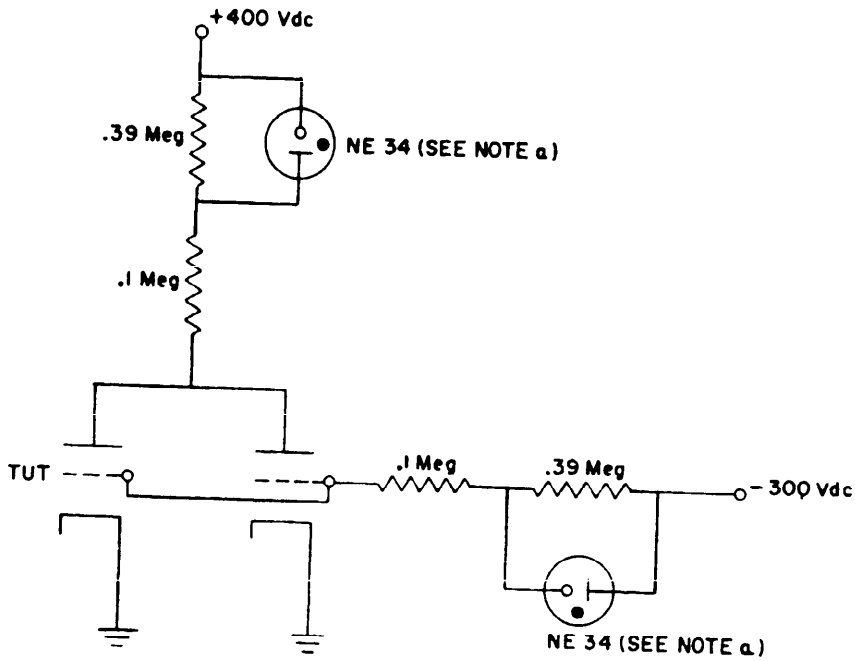


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	----	4.750 (120.65)
B	----	2.070 (52.58)
C	----	4.175 (106.05)
Reference dimension (see note a)		
D	1.750 (44.45)	

NOTE:
a. Dimensions without tolerances are for information and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 6528A.

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NOTE:
a. Type NE 34 lamp shall conform to MIL-L-15098.

FIGURE 2. Tap shorts circuit.

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